Normal Distribution

ClipSAT Printable Practice Suite (clipsat.org) • Topic 19.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Properties of the normal distribution**

State two key properties of the normal distribution curve.

The normal distribution curve is symmetric (bell-shaped) about its mean μ, and the total area under the curve equals 1. (Other valid properties: mean = median = mode; the curve is asymptotic to the horizontal axis.)

2 **[Foundational]** • **Target: Properties of the normal distribution**

Explain what it means for the normal distribution curve to be "asymptotic" to the horizontal axis.

It means the curve approaches the horizontal axis as $x \to \pm\infty$ but never actually touches or crosses it — the probability density gets arbitrarily close to zero but is never exactly zero for any finite x.

3 **[Foundational]** • **Target: Properties of the normal distribution**

State one similarity and one difference between the normal distribution and the discrete uniform distribution.

Similarity: both are probability distributions in which the total probability sums (or integrates) to 1. Difference: the normal distribution is continuous and bell-shaped, with values clustering near the mean, while the discrete uniform distribution is discrete and every outcome has equal probability.

4 **[Foundational]** • **Target: Standardizing a normal variable**

A random variable $X \sim N(50, 64)$. Find the z-score corresponding to $x = 62$, using $z = \frac{x-\mu}{\sigma}$.

$\sigma = \sqrt{64} = 8$. $z = \frac{62-50}{8} = \frac{12}{8} = 1.5$.

5 **[Foundational]** • **Target: Standardizing a normal variable**

A random variable $X \sim N(30, 25)$. Find the z-score corresponding to $x = 23$.

$\sigma = \sqrt{25} = 5$. $z = \frac{23-30}{5} = \frac{-7}{5} = -1.4$.

6 **[Foundational]** • **Target: Finding probabilities using the standard normal table**

A random variable $X \sim N(100, 225)$. Find $P(X < 115)$, given that $\Phi(1) \approx 0.8413$.

$\sigma = \sqrt{225} = 15$. $z = \frac{115-100}{15} = 1$. $P(X < 115) = \Phi(1) \approx 0.8413$.

© ClipSAT — clipsat.org | All Rights Reserved

7 [Foundational] • Target: Finding probabilities using the standard normal table

Using the same distribution $X \sim N(100, 225)$, find $P(X > 130)$, given that $\Phi(2) \approx 0.9772$.

$z = \frac{130 - 100}{15} = 2$. $P(X > 130) = 1 - \Phi(2) \approx 1 - 0.9772 = 0.0228$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding a value given a probability (inverse normal)

A random variable $X \sim N(60, 100)$. Find the value x such that $P(X < x) = 0.9332$, given that $\Phi(1.5) \approx 0.9332$.

Since $\Phi(1.5) \approx 0.9332$, the corresponding z-value is 1.5. $x = \mu + z\sigma = 60 + 1.5(10) = 75$.

9 [Intermediate] • Target: The normal approximation to the binomial distribution

A random variable $X \sim B(200, 0.4)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid (both np and $n(1-p)$ should exceed 5).

$np = 200(0.4) = 80$ and $n(1-p) = 200(0.6) = 120$, both exceed 5, so the approximation is valid. Use $X \approx N(80, 48)$, since $\text{Var}(X) = np(1-p) = 200(0.4)(0.6) = 48$.

10 [Intermediate] • Target: The normal approximation to the binomial distribution

A random variable $X \sim B(150, 0.5)$ is to be approximated by a normal distribution. State the mean and variance to use, and check the approximation is valid.

$np = 150(0.5) = 75$ and $n(1-p) = 150(0.5) = 75$, both exceed 5, so the approximation is valid. Use $X \approx N(75, 37.5)$, since $\text{Var}(X) = np(1-p) = 150(0.5)(0.5) = 37.5$.

11 [Intermediate] • Target: Solving for unknown parameters of a normal distribution

A random variable $X \sim N(\mu, 36)$ satisfies $P(X < 50) = 0.8413$, given $\Phi(1) \approx 0.8413$. Find μ.

$\sigma = 6$, and the z-value corresponding to probability 0.8413 is $z = 1$. So $\frac{50 - \mu}{6} = 1$, giving $\mu = 50 - 6 = 44$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: A well-designed word problem: setting a pass mark for an exam

This question has two parts. Scores on a standardized exam are normally distributed with mean $\mu = 68$ and standard deviation $\sigma = 12$. The exam board wants to set a pass mark so that exactly the top 10% of students pass, and is told that the z-value corresponding to the top 10% (i.e. $P(Z > z) = 0.10$) is $z \approx 1.2816$. (a) Find the pass mark, to the nearest whole number. (b) A student scores 90 on the exam. Find their z-score, and state whether they are in the top 10%.

(a) $x = \mu + z\sigma = 68 + 1.2816(12) \approx 68 + 15.38 \approx 83$ (to the nearest whole number). (b) $z = \frac{90 - 68}{12} = \frac{22}{12} \approx 1.833$. Since a score of 90 exceeds the pass mark of approximately 83 (and $z \approx 1.833 > 1.2816$), this student IS in the top 10%.

13 [Intermediate] • Target: The probability of a value lying between two bounds

Practice 3.2: Ship A leaves port P on a bearing of 040° traveling at 15 km/h. At the same time, ship B leaves P on a bearing of 130° traveling at 20 km/h. Find the distance between the two ships after 2 hours.

1. Distance travelled by ship A: $d_A = 15 \times 2 = 30$ km. 2. Distance travelled by ship B: $d_B = 20 \times 2 = 40$ km. 3. The angle between their paths is the difference in bearings: $\theta = 130^\circ - 40^\circ = 90^\circ$. 4. Since the triangle is right-angled at P, apply the Pythagorean theorem (or Cosine Rule with $\cos 90^\circ = 0$): $d = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km. The distance between the ships after 2 hours is exactly 50 km.

14 [Intermediate] • Target: The normal approximation to the binomial distribution

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

(a) $P(X = 5) = \frac{e^{-3} \cdot 243}{120} \approx 0.1008$. (b) $\lambda = 1.5$: $P(X \geq 2) = 1 - P(0) - P(1) = 1 - e^{-1.5} - 1.5e^{-1.5} = 1 - 2.5e^{-1.5} \approx 0.4422$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 15.01 Representation of Data

[Poisson Distribution · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging]** • **Target: A well-designed word problem: setting a pass mark for an exam** • **Cross-Lesson: 16.01 Permutations & Combinations**

[Polynomials, Remainder Theorem & Binomial Theorem · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

17 **[Challenging]** • **Target: Properties of the normal distribution** • **Cross-Lesson: 17.01 Discrete Random Variables**

[Vectors in Two Dimensions & Relative Motion · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

18 **[Challenging]** • **Target: Standardizing a normal variable** • **Cross-Lesson: 18.01 Binomial Distribution**

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 **[Challenging]** • **Target: Finding probabilities using the standard normal table** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

In a factory, 60% of components are manufactured by Machine A and 40% by Machine B. It is known that 3% of components produced by Machine A are defective, whereas 5% of components produced by Machine B are defective. (a) Find the probability that a randomly chosen component is defective. (b) Given that a randomly chosen component is defective, find the probability that it was produced by Machine B. (c) Determine whether the events "component produced by Machine B" and "component is defective" are independent.

(a) By the law of total probability: $P(D) = P(A)P(D \mid A) + P(B)P(D \mid B) = (0.60)(0.03) + (0.40)(0.05)$ $= 0.018 + 0.020 = 0.038 \quad (3.8\%)$. (b) By Bayes' Theorem / conditional probability: $P(B \mid D) = \frac{P(B \cap D)}{P(D)} = \frac{P(B)P(D \mid B)}{P(D)} = \frac{(0.40)(0.05)}{0.038} = \frac{0.020}{0.038} = \frac{20}{38} = \frac{10}{19} \approx 0.5263 \quad (52.6\%)$. (c) Check the independence condition $P(B \cap D) = P(B)P(D)$: $P(B \cap D) = 0.020$. $P(B) \times P(D) = 0.40 \times 0.038 = 0.0152$..

20 **[Challenging]** • **Target: Finding a value given a probability (inverse normal)** • **Cross-Lesson: 3.01 Graphs & Transformations**

Given vectors $\mathbf{a} = 2\mathbf{i} - 5\mathbf{j}$ and $\mathbf{b} = -3\mathbf{i} + 4\mathbf{j}$, find the scalar projection of $\mathbf{a}$ onto $\mathbf{b}$ and the angle θ between them to the nearest tenth of a degree.

Calculate the dot product: $\mathbf{a} \cdot \mathbf{b} = (2)(-3) + (-5)(4) = -6 - 20 = -26$. Calculate the magnitudes: $|\mathbf{a}| = \sqrt{2^2 + (-5)^2} = \sqrt{4 + 25} = \sqrt{29} \approx 5.385$, $|\mathbf{b}| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Scalar projection of $\mathbf{a}$ onto $\mathbf{b}$: $\text{comp}_{\mathbf{b}}\mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|} = \frac{-26}{5} = -5.2$. Angle θ between vectors.

© ClipSAT — clipsat.org | All Rights Reserved

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Properties of the normal distribution

The histogram below shows the number of hours 100 students spent studying for an exam. The number of students who studied for between 0 and 5 hours is represented by a bar with a height of 30. Determine the number of students who studied for between 5 and 10 hours.

a. **20** ✓

b. 25

c. 30

d. 35

The histogram displays the distribution of study hours for 100 students. The total number of students is 100. The question states that the number of students who studied for between 0 and 5 hours is 30. The bar for the '0-5 hours' interval has a height of 30, representing 30 students. The bars for the '5-10 hours' and '10-15 hours' intervals have heights of 20 and 15 respectively, representing 20 and 15 students. Therefore, the number of students who studied for between 5 and 10 hours is 20.

22 [Foundational] • Target: Standardizing a normal variable

The following two-way table shows the results of a survey on favourite sports among a group of students. Calculate the total number of students surveyed.

a. 50

b. 45

c. **60** ✓

d. 55

The total number of students surveyed is the sum of all entries in the two-way table.
$\text{Total} = 15 + 10 + 20 + 15 = 60$.

23 [Foundational] • Target: Finding probabilities using the standard normal table

Normal distribution $X \sim N(50, 9)$: P(X<50)=

a. 0

b. 0.25

c. **0.5** ✓

d. 1

Mean=50, so P(X<50)=0.5 by symmetry.

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Finding a value given a probability (inverse normal)

Two circles, C_1 with centre O_1 and radius 6 cm, and C_2 with centre O_2 and radius $6\sqrt{3}$ cm, intersect at the points A and B. The distance between their centres is $O_1O_2 = 12$ cm. Find the exact area of the region common to both circles, in cm^2.

a. $30\pi - 36\sqrt{3}$ ✓

b. $30\pi + 36\sqrt{3}$

c. $42\pi - 36\sqrt{3}$

d. $15\pi - 18\sqrt{3}$

1. **Determine the angles subtended at each centre:** Consider triangle O_1AO_2 with side lengths $O_1A = 6$, $O_2A = 6\sqrt{3}$, and $O_1O_2 = 12$. Notice that: $(O_1A)^2 + (O_2A)^2 = 6^2 + (6\sqrt{3})^2 = 36 + 108 = 144 = 12^2 = (O_1O_2)^2$ By the converse of Pythagoras' Theorem, $\angle O_1AO_2 = \frac{\pi}{2}$ radians. Using right-angled trigonometry in ΔO_1AO_2.

25 [Foundational] • Target: The probability of a value lying between two bounds

A straight line passing through the origin is tangent to both the curve C_1 with equation $y = e^{kx}$, where k is a positive constant, and the curve C_2 with equation $y = \ln x$. Determine the exact distance between the two points of contact.

a. $(e-1)\sqrt{e^2+1}$ ✓

b. $e\sqrt{e^2+1}$

c. $(e-1)\sqrt{e^2-1}$

d. $\sqrt{e^4-e^2+1}$

We synthesize differentiation with coordinate geometry to find the two points of contact. **Step 1: Point of tangency on $C_2 : y = \ln x$** Let $A(x_1, y_1)$ be the point of contact on C_2, so $y_1 = \ln x_1$. The derivative is $\frac{dy}{dx} = \frac{1}{x}$, so the gradient of the tangent at A is $\frac{1}{x_1}$. Since the tangent line passes through the origin $(0, 0)$, its gradient is also: $m = \frac{y_1 - 0}{x_1 - 0} = \frac{\ln x_1}{x_1}$ Equating the two expressions for the gradient.

26 [Foundational] • Target: The normal approximation to the binomial distribution

In a Voronoi diagram, two sites are located at $A(2, 4)$ and $B(6, 2)$. Which of the following is the equation of the boundary edge lying on the perpendicular bisector of the line segment AB?

a. $y = 2x - 5$ ✓

b. $y = -\frac{1}{2}x + 5$

c. $y = 2x - 8$

d. $y = -2x + 11$

In a Voronoi diagram, the boundary edge between two sites is along the perpendicular bisector of the line segment connecting them. 1. Find the midpoint M of the segment AB: $M = \left(\frac{2+6}{2}, \frac{4+2}{2}\right) = (4, 3)$ 2. Find the gradient of segment AB: $m_{AB} = \frac{2-4}{6-2} = \frac{-2}{4} = -\frac{1}{2}$ 3. The gradient of the perpendicular bisector is the negative reciprocal of m_{AB}: $m_\perp = -\frac{1}{-\frac{1}{2}} = 2$ 4. Use the point-slope form with midpoint $(4, 3)$ and.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Solving for unknown parameters of a normal distribution

In triangle ABC, point D lies on side BC such that AD is an angle bisector of angle A. Let $AB = c$, $AC = b$, $BD = m$, and $CD = n$. Prove that $\frac{m}{n} = \frac{c}{b}$ using a combination of the Sine Rule and the properties of angle bisectors.

a. This statement is a direct consequence of the Sine Rule applied to triangles ABD and ACD. ✓

b. This statement can be proven by applying the Law of Cosines to triangles ABD and ACD and equating expressions for AD^2.

c. This statement is true only if triangle ABC is isosceles with $b = c$.

d. This statement requires the use of vector methods for its proof.

Let $\angle BAD = \angle CAD = \alpha$. In triangle ABD, by the Sine Rule, we have $\frac{m}{\sin \alpha} = \frac{c}{\sin \angle ADB}$. Therefore, $m = \frac{c \sin \alpha}{\sin \angle ADB}$. In triangle ACD, by the Sine Rule, we have $\frac{n}{\sin \alpha} = \frac{b}{\sin \angle ADC}$. Therefore, $n = \frac{b \sin \alpha}{\sin \angle ADC}$. Since $\angle ADB$ and $\angle ADC$ are supplementary angles, $\sin \angle ADB = \sin(180^\circ - \angle ADC) = \sin \angle ADC$. Now, we can form the ratio.

28 [Foundational] • Target: A well-designed word problem: setting a pass mark for an exam

The histogram below shows the distribution of scores for a class of students on a recent quiz. The total number of students who took the quiz is 50. The frequency for the score interval [70, 80) is x. Determine the value of x.

a. 5

b. 10 ✓

c. 15

d. 20

The histogram represents the frequency distribution of quiz scores. The total number of students is given as 50. The frequencies for the score intervals are as follows: [50, 60): 8 [60, 70): 12 [70, 80): x [80, 90): 7 [90, 100]: 8 The sum of the frequencies must equal the total number of students: 8 + 12 + x + 7 + 8 = 50 Combine the known frequencies: 35 + x = 50 Solve for x: x = 50 - 35 x = 15 Therefore, the frequency for the score interval [70, 80) is 15.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: Properties of the normal distribution

Find the standard deviation of the data set: $12, 19, 14, 22, 24$, correct to 2 decimal places.

a. 4.98

b. **4.58** ✓

c. 4.18

d. 3.78

Working through this statistics step: Mean $= \frac{12+19+14+22+24}{5} = 18.20$. $\sigma = \sqrt{\frac{\sum(x-\bar{x})^2}{n}} \approx 4.58$.

30 [Intermediate] • Target: Standardizing a normal variable

A bag contains 8 red and 9 blue balls. Two balls are drawn at random without replacement. Find $P(\text{both red})$.

a. $\frac{64}{289}$

b. $\frac{8}{17}$

c. $\frac{7}{34}$ ✓

d. $\frac{56}{289}$

Working through this statistics step: $P(\text{both red}) = \frac{8}{17} \times \frac{7}{16} = \frac{7}{34}$.

31 [Intermediate] • Target: Finding probabilities using the standard normal table

Solve $|x - 6| > 6$. What is the smaller boundary value separating the two regions?

a. 11

b. **0** ✓

c. 12

d. 1

$x - 6 > 6$ or $x - 6 < -6$, giving boundary values $x = 12$ and $x = 0$. Smaller boundary: $x = 0$.

32 [Intermediate] • Target: Finding a value given a probability (inverse normal)

Solve $|2x + 3| = 3$ for x. What is the larger of the two solutions?

a. **0** ✓

b. -3

c. 3

d. $\frac{1}{12}$

$2x + 3 = \pm 3$, giving $x = 0$ or $x = -3$. The larger solution is $x = 0$.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: The probability of a value lying between two bounds

Two lines have direction vectors $\mathbf{d_1} = (-2, 3, -5)$ and $\mathbf{d_2} = (4, -6, 10)$. What is the relationship between the two lines (assuming they are coplanar)?

a. neither parallel nor perpendicular

b. perpendicular

c. identical

d. **parallel ✓**

$\mathbf{d_2} = -2\mathbf{d_1}$, so the lines are parallel.

34 [Intermediate] • Target: The normal approximation to the binomial distribution

Expand $\log\left(\frac{x}{y}\right)$ using properties of logarithms. (variant 6)

a. $\frac{\log x}{\log y}$

b. $\log(x) \cdot \log(y)$

c. $\log x + \log y$

d. $\log x - \log y$ ✓

Quotient rule: $\log(x/y) = \log x - \log y$.

35 [Intermediate] • Target: Solving for unknown parameters of a normal distribution

Expand $\log\left(\frac{x^2}{y^3}\right)$ using properties of logarithms.

a. $\log(x) \cdot \log(y)$

b. $2\log x - 3\log y$ ✓

c. $2\log x + 3\log y$

d. $\frac{\log x}{\log y}$

Combine power and quotient rules.

36 [Intermediate] • Target: A well-designed word problem: setting a pass mark for an exam

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.

a. (3, 0)

b. (1, 2)

c. **(2, 1)** ✓

d. (4, -1)

The graph of $f(x) = \log_2(x)$ intersects the x-axis when $f(x) = 0$. Thus, $\log_2(x) = 0$, which implies $x = 2^0 = 1$. So, point Q has coordinates (1, 0). The line segment QR is perpendicular to the x-axis, meaning that the x-coordinate of R is the same as the x-coordinate of Q. Therefore, the x-coordinate of R is 1. Point R lies on the graph of $g(x) = 3 - x$. Substituting the x-coordinate of R into the equation for $g(x)$, we get $g(1) = 3 - 1 = 2$. Thus, the coordinates of point R are (1, 2). Let's check the.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Properties of the normal distribution

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$. Find the area of the region enclosed between the curve and the x-axis.

a. 27

b. **36** ✓

c. 54

d. 72

To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$: Area $= \int_0^6 (6x - x^2)\,dx$ Integrate term by term: $\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$ Substitute the limits: $\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$ Common errors: - Estimating the region as a triangle with base 6 and maxim.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Standardizing a normal variable • Cross-Lesson: 18.01 Binomial Distribution

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.

a. 3

b. 4

c. **5** ✓

d. 6

To find the intersection points of the two functions, we set $f(x) = g(x)$. $x^2 - 6x + 11 = 2x - 1$ Rearrange the equation to form a quadratic equation: $x^2 - 6x - 2x + 11 + 1 = 0$ $x^2 - 8x + 12 = 0$ We can factor this quadratic equation. We are looking for two numbers that multiply to 12 and add to -8. These numbers are -6 and -2. $(x - 2)(x - 6) = 0$ This gives us two possible solutions for x: $x = 2$ or $x = 6$. We are given that one intersection point has an x-coordinate of 2. Therefore, the x-coordinate of.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Finding probabilities using the standard normal table • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Consider the quadratic function $f(x) = x^2 - 4x + 3$. The graph of this function is shown below. The points where the graph intersects the x-axis are denoted by A and B. What is the distance between points A and B?

a. 1

b. 2

c. **3** ✓

d. 4

The points where the graph intersects the x-axis are the roots of the quadratic equation $f(x) = 0$. We need to solve $x^2 - 4x + 3 = 0$. Factoring the quadratic, we get $(x - 1)(x - 3) = 0$. Thus, the roots are $x = 1$ and $x = 3$. Let point A have coordinates $(1, 0)$ and point B have coordinates $(3, 0)$. The distance between A and B is the absolute difference of their x-coordinates: $|3 - 1| = 2$.

40 [Challenging] • Target: Finding a value given a probability (inverse normal) • Cross-Lesson: 3.01 Graphs & Transformations

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

a. **(1, 0)** ✓

b. (3, 0)

c. (0, 3)

d. (0, 1)

To find the x-intercepts of the function $f(x) = x^2 - 4x + 3$, we set $f(x) = 0$. This gives us the quadratic equation $x^2 - 4x + 3 = 0$. We can factor this equation as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. Therefore, the x-intercepts are at $(1, 0)$ and $(3, 0)$. The point with the smaller x-value is $(1, 0)$.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: The probability of a value lying between two bounds • Cross-Lesson: 4.01 Coordinate Geometry

A curve has equation $y = x\sqrt{16 - x^2}$ for $0 \le x \le 4$. A point $P(x, y)$ moves along the curve in the first quadrant, and the normal to the curve at P intersects the x-axis at the point N. Given that the x-coordinate of P is increasing at a constant rate of $\sqrt{3}$ units s^{-1}, find the rate of change of the area of triangle OPN, where O is the origin, at the instant when $x = 2$.

a. -6 units^2 s^{-1} ✓

b. 6 units^2 s^{-1}

c. -12 units^2 s^{-1}

d. $-2\sqrt{3}$ units^2 s^{-1}

Step 1: Differentiate $y = x\sqrt{16 - x^2}$ using the product rule: $\frac{dy}{dx} = 1 \cdot \sqrt{16 - x^2} + x \cdot \frac{-2x}{2\sqrt{16 - x^2}} = \frac{16 - 2x^2}{\sqrt{16 - x^2}}$ At $x = 2$: $y = 2\sqrt{16 - 4} = 4\sqrt{3}$ $\frac{dy}{dx} = \frac{16 - 2(4)}{\sqrt{12}} = \frac{8}{2\sqrt{3}} = \frac{4}{\sqrt{3}}$ Step 2: Find the coordinates of N, the x-intercept of the normal. The gradient of the normal at P is: $m_N = -\frac{1}{\frac{dy}{dx}} = -\frac{\sqrt{16 - x^2}}{16 - 2x^2}$ The equation of the no.

42 [Challenging] • Target: The normal approximation to the binomial distribution • Cross-Lesson: 5.01 Trigonometry

The curve C has equation $y = x^3 - 3x + 2$. The tangent to C at the point $P(2, 4)$ intersects the curve again at point Q. Find the area of the finite region completely enclosed between the curve C and the tangent line.

a. **108** ✓

b. 84

c. 64

d. 12

To find the area of the enclosed region, we synthesize differential calculus, polynomial algebra, and definite integration: 1. **Find the equation of the tangent line at $P(2, 4)$:** Differentiating the curve equation: $\frac{dy}{dx} = 3x^2 - 3$ At $x = 2$, the gradient of the tangent is: $m = 3(2)^2 - 3 = 12 - 3 = 9$ Using the point-slope formula with $P(2, 4)$: $y - 4 = 9(x - 2) \Rightarrow y = 9x - 14$ 2. **Find the intersection point Q:** Set the curve and tangent equations equal.

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Solving for unknown parameters of a normal distribution • Cross-Lesson: 6.01 Differentiation & Integration

Consider the two functions $f(x) = ax^2 + bx + c$ and $g(x) = dx + e$. The graph of $f(x)$ is a parabola with its vertex at $(3, -1)$ and passes through the point $(1, 3)$. The graph of $g(x)$ is a straight line passing through the points $(2, 5)$ and $(6, -3)$. If the graphs of $f(x)$ and $g(x)$ intersect at two points, and the x-coordinate of one of these intersection points is p, find the value of $\frac{a+d}{p}$ given that p is an integer.

a. 3

b. **2** ✓

c. -2

d. -3

First, we find the equation for $f(x)$. Since the vertex is at $(3, -1)$, we can write $f(x) = a(x-3)^2 - 1$. Given that the graph passes through $(1, 3)$: $3 = a(1-3)^2 - 1$ $3 = a(-2)^2 - 1$ $3 = 4a - 1$ $4 = 4a$ $a = 1$. So, $f(x) = 1(x-3)^2 - 1 = x^2 - 6x + 9 - 1 = x^2 - 6x + 8$. Thus, $a = 1$. Next, we find the equation for $g(x)$. The slope d is $\frac{3-2}{1-0} = \frac{1}{1} = 1$. Using the point-slope form with $(0, 2)$: $y - 2 = 1(x - 0)$ $y = x + 2$. So, $g(x) = x + 2$. Thus, $d = 1$. Now, we find t.

44 [Challenging] • Target: A well-designed word problem: setting a pass mark for an exam • Cross-Lesson: 7.01 Statistics & Probability

Two parallel lines, L_1 and L_2, have n and $n + 2$ distinct points marked on them respectively, where n is an integer with $n \geq 2$. Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is $6 : 5$, find the total number of triangles that can be formed.

a. 175 ✓

b. 210

c. 220

d. 96

1. **Determine the number of triangles, T:** Three points form a triangle if and only if they are not all on the same straight line. - Case 1: 1 point from L_1 and 2 points from L_2: $\binom{n}{1}\binom{n+2}{2} = n \cdot \frac{(n+2)(n+1)}{2}$ - Case 2: 2 points from L_1 and 1 point from L_2: $\binom{n}{2}\binom{n+2}{1} = \frac{n(n-1)}{2} \cdot (n+2)$ Summing both cases: $T = \frac{n(n+2)}{2}[(n+1) + (n-1)] = \frac{n(n+2)}{2}(2n) = n^2(n+2)$ 2. **Determine the number of quadrilaterals, Q.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Properties of the normal distribution • Cross-Lesson: 8.01 Functions: Composite & Inverse

The region R in the first quadrant is bounded by the curve $y = \frac{6}{\sqrt{3x+4}}$, the coordinate axes, and the vertical line $x = k$, where $k > 0$. Given that the total area of R is 8, and the vertical line $x = c$ (where $0 < c < k$) divides R into two regions of equal area, find the exact value of c.

a. $\frac{5}{3}$ ✓

b. 2

c. $\frac{7}{3}$

d. $\frac{13}{27}$

Step 1: Integrate the function using the reverse chain rule. $\int 6(3x+4)^{-\frac{1}{2}}\,dx = \frac{6}{\frac{1}{2}\cdot 3}(3x+4)^{\frac{1}{2}} = 4\sqrt{3x+4}$

Step 2: Find the parameter k using the given total area of 8.

$\text{Area} = \int_0^k \frac{6}{\sqrt{3x+4}}\,dx = \left[4\sqrt{3x+4}\right]_0^k = 4\sqrt{3k+4} - 4\sqrt{4} = 4\sqrt{3k+4} - 8$ Setting this equal to 8:

$4\sqrt{3k+4} - 8 = 8 \Rightarrow 4\sqrt{3k+4} = 16 \Rightarrow \sqrt{3k+4} = 4$ Squaring both.

© ClipSAT — clipsat.org | All Rights Reserved